Generative Modelling with Stochastic Differential Equations

Exercises, week 1

Sampling and the score

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Exercises, chapters and parts marked ($\star$) are on material that is not covered in the lectures and is left for self-study. It is examinable like the rest.

1 Sampling in high dimension

Exercise 1.1 (Bayesian logistic regression)**.** Let $(a_i, y_i) \in \mathbb{R}^n \times \{-1, +1\}$, $i = 1, \dots, N$, and $\lambda > 0$, and consider the potential

$$U(\theta) := \sum_{i=1}^{N} \ell(y_i\, a_i^\top \theta) + \frac{\lambda}{2}|\theta|^2, \qquad \ell(s) := \log(1 + e^{-s}), \qquad \theta \in \mathbb{R}^n.$$

(a) Show that e^{-U} is, up to a constant, the posterior density of θ in the model $\mathbb{P}(y_i = y \mid a_i, \theta) = (1 + e^{-y\, a_i^\top \theta})^{-1}$, $y \in \{-1, +1\}$, with independent labels and prior $\theta \sim \mathcal{N}(0, \lambda^{-1}\mathbb{I}_n)$.

(b) Compute ∇U and $\nabla^2 U$, and show that $0 < \ell'' \le \frac{1}{4}$.

(c) Deduce that $\alpha\, \mathbb{I}_n \preceq \nabla^2 U(\theta) \preceq \beta\, \mathbb{I}_n$ for all θ, with $\alpha = \lambda$ and $\beta = \lambda + \frac{1}{4}\lambda_{\max}\big(\sum_i a_i a_i^\top\big) \le \lambda + \frac{1}{4}\sum_i |a_i|^2$.

(d) Show that $Z = \int e^{-U}$ is finite, and that U has a unique minimiser $\theta^\star$ (the maximum a posteriori estimator).

Exercise 1.2 (Generalised inverse CDF)**.** Let μ be a probability measure on $\mathbb{R}$ with CDF $F(t) = \mu((-\infty, t])$, and define the generalised inverse

$$F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \ge u\}, \qquad u \in (0, 1).$$

(a) Show that $F^{-1}(u) \in \mathbb{R}$ and that, for all $u \in (0, 1)$ and $t \in \mathbb{R}$,

$$F^{-1}(u) \le t \iff u \le F(t).$$

(b) Deduce that $X = F^{-1}(U)$ with $U \sim \mathrm{Unif}[0, 1]$ has law μ. No continuity or invertibility of F is needed.

(c) Compute F^{-1} for $\mu = \mathrm{Exp}(\lambda)$ and for $\mu = \mathrm{Bernoulli}(p)$.

(d) Let $I_1, \dots, I_m \subset \mathbb{R}^n$ be disjoint bounded measurable sets of positive volume (cells) and $\hat{p}_1, \dots, \hat{p}_m \ge 0$ with $\sum_i \hat{p}_i = 1$, and set $F_i = \sum_{k \le i} \hat{p}_k$. The histogram sampler draws $U \sim \mathrm{Unif}[0, 1]$, finds the smallest i with $U \le F_i$, and returns a uniform point of I_i. Show that its output has the histogram density $\sum_{i=1}^m \hat{p}_i\, \mathbf{1}_{I_i}/|I_i|$.

Hint for (a): show that $\{x : F(x) \ge u\}$ is a closed half-line $[x_u, \infty)$.

Rejection sampling. Exercises 1.3–1.5 use the following setting. The target is a probability density $\mu = \tilde{\mu}/Z_\mu$ on $\mathbb{R}^n$, known only through an unnormalized density $\tilde{\mu} \ge 0$ with $Z_\mu := \int \tilde{\mu} \in (0, \infty)$. The proposal is a density $\nu = \tilde{\nu}/Z_\nu$ that we can sample from, where $Z_\nu := \int \tilde{\nu} < \infty$ and $\tilde{\nu} \ge \tilde{\mu}$ everywhere ($\tilde{\nu}$ is an *upper envelope* of $\tilde{\mu}$). Rejection sampling draws $X \sim \nu$ and an independent $U \sim \mathrm{Unif}[0, 1]$, and accepts X if $U \le \tilde{\mu}(X)/\tilde{\nu}(X)$. Otherwise it repeats with a new independent pair. The output is the first accepted proposal. The *acceptance probability* is the probability that a single proposal is accepted.

Exercise 1.3 (The number of proposals in rejection sampling)**.** In the rejection sampling setting above, let $(X_k)_{k \ge 1} \overset{\text{i.i.d.}}{\sim} \nu$ and $(U_k)_{k \ge 1} \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}[0, 1]$ be independent, let $r = \tilde{\mu}/\tilde{\nu} \in [0, 1]$, and let $T = \inf\{k \ge 1 : U_k \le r(X_k)\}$ be the index of the first accepted proposal.

(a) Show that T is geometric with success probability $p = Z_\mu/Z_\nu$, i.e. $\mathbb{P}(T = k) = (1-p)^{k-1}p$, so that $T < \infty$ almost surely and $\mathbb{E}T = Z_\nu/Z_\mu$.

(b) Show that $X_T \sim \mu$ and that X_T is independent of T.

(c) Explain why the algorithm can be run without knowing Z_μ, and what happens if the envelope condition $\tilde\nu \geq \tilde\mu$ fails on a set of positive ν-measure.

Exercise 1.4 (Rejection sampling of a Gaussian)**.** Let the target be $\mu = \mathcal{N}(0, \mathbb{I}_n)$ with $\tilde\mu(x) = e^{-|x|^2/2}$, and the proposal $\nu = \mathcal{N}(0, s^2\mathbb{I}_n)$ for some $s > 0$, with envelopes of the form $\tilde\nu_c(x) = c\, e^{-|x|^2/(2s^2)}$, $c > 0$.

(a) Show that an envelope $\tilde\nu_c \geq \tilde\mu$ exists if and only if $s \geq 1$, and that the smallest valid c is then $c = 1$.

(b) Show that the acceptance probability with $c = 1$ is s^{-n}.

(c) With $s = 1.01$, how many proposals are needed on average when $n = 1000$?

(d) Now let the target be $\mathcal{N}(m, \mathbb{I}_n)$ and keep the proposal $\mathcal{N}(0, s^2\mathbb{I}_n)$, $s > 1$. Show that the best acceptance probability is $s^{-n}\exp\left(-\frac{|m|^2}{2(s^2-1)}\right)$.

Exercise 1.5 (Volume of the unit ball)**.** Let $\mathcal{B}_2^n = \{x \in \mathbb{R}^n : |x| \leq 1\}$ and let $\omega_n := |\mathcal{B}_2^n|$ be its volume.

(a) Using $\int_{\mathbb{R}} e^{-t^2}\mathrm{d}t = \sqrt{\pi}$ and the layer-cake formula $\int_{\mathbb{R}^n} \varphi\,\mathrm{d}x = \int_0^\infty |\{\varphi > t\}|\,\mathrm{d}t$ for $\varphi \geq 0$, compute $\int_{\mathbb{R}^n} e^{-|x|^2}\mathrm{d}x$ in two ways and deduce

$$\omega_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}.$$

Check that for odd n this equals $2(2\pi)^{(n-1)/2}/n!!$, but that this last formula is wrong for even n.

(b) Show that $\omega_{n+2} = \frac{2\pi}{n+2}\,\omega_n$, and deduce that $\omega_n < 1$ if and only if $n \geq 13$. Consider rejection sampling of the uniform distribution on $\mathcal{B}_2^n$ with $\tilde\mu = \mathbf{1}_{\mathcal{B}_2^n}$ and envelope $\tilde\nu = \mathbf{1}_{[-1,1]^n}$. Show that its acceptance probability is $\omega_n/2^n$ and conclude that it is smaller than 2^{-n} if and only if $n \geq 13$.

(c) Using Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\,(x/e)^x$, show that $\omega_n 2^{-n} \sim \frac{1}{\sqrt{\pi n}}\left(\frac{\pi e}{2n}\right)^{n/2}$.

Exercise 1.6 (Maximum likelihood is KL minimisation)**.** Recall the Kullback–Leibler divergence $\mathrm{KL}(\mu\|g) := \int \mu\log(\mu/g)$ (a sum on a finite set), with $\mathrm{KL}(\mu\|g) = +\infty$ if μ is not absolutely continuous with respect to g, and the maximum likelihood estimator $\arg\max_{g\in\mathcal{G}} \frac{1}{N}\sum_{i=1}^N \log g(x_i)$ over a family $\mathcal{G}$ of densities.

(a) Let μ and g be probability densities on $\mathbb{R}^n$ with $\mathbb{E}_\mu|\log\mu| < \infty$ and $\mathbb{E}_\mu|\log g| < \infty$. Show that $\mathbb{E}_{X\sim\mu}\log g(X) = \mathbb{E}_\mu\log\mu - \mathrm{KL}(\mu\|g)$, and that $\mathrm{KL}(\mu\|g) \geq 0$ with equality iff $g = \mu$ a.e. Deduce that, over a family $\mathcal{G}$, $\arg\max_{g\in\mathcal{G}} \mathbb{E}_\mu\log g = \arg\min_{g\in\mathcal{G}} \mathrm{KL}(\mu\|g)$, which contains μ if $\mu \in \mathcal{G}$.

(b) Let μ be a distribution on a finite set and $\hat\mu_N = \frac{1}{N}\sum_i \delta_{x_i}$ the empirical measure. Show that the MLE over a family $\mathcal{G}$ of probability mass functions is exactly the minimiser of $g \mapsto \mathrm{KL}(\hat\mu_N\|g)$ over $\mathcal{G}$. Why does this fail for densities on $\mathbb{R}^n$?

(c) Let $\mathcal{G} = \{\mathcal{N}(m, \Sigma) : m \in \mathbb{R}^n, \Sigma \succ 0\}$ and let μ be a probability density on $\mathbb{R}^n$ with $\mathbb{E}_\mu |\log \mu| < \infty$, finite second moments, mean m_μ and covariance $\Sigma_\mu \succ 0$. Show that $\arg\min_{g \in \mathcal{G}} \mathrm{KL}(\mu \| g) = \mathcal{N}(m_\mu, \Sigma_\mu)$. Deduce that, when the empirical covariance of $x_1, \ldots, x_N$ is positive definite, the Gaussian MLE is the Gaussian with the empirical mean and covariance.

2 Langevin dynamics and the score

Exercise 2.1 (ULA on a Gaussian target)**.** Let $U(x) = \frac{1}{2}\langle x, Ax\rangle$, where A is a symmetric $n \times n$ matrix with eigenvalues $0 < \lambda_{\min} = \lambda_1 \le \cdots \le \lambda_n = \lambda_{\max}$, so that $\mu = \mathcal{N}(0, A^{-1})$, and consider ULA with step size $h > 0$,

$$X_{k+1} = (\mathbb{I}_n - hA)X_k + \sqrt{2h}\,\xi_k, \qquad \xi_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbb{I}_n),$$

with X_0 independent of $(\xi_k)_k$. Example 2.4 treats the isotropic case $A = \alpha\mathbb{I}_n$.

(a) Show that ULA has a stationary distribution if and only if $h < 2/\lambda_{\max}$, that it is then unique, and that it equals $\mu_h = \mathcal{N}(0, (A(\mathbb{I}_n - hA/2))^{-1})$.

(b) In the isotropic case $A = \alpha\mathbb{I}_n$, prove the expansions of $\mathcal{W}_2(\mu_h, \mu)$ and $\mathrm{KL}(\mu_h\|\mu)$ stated in Example 2.4.

(c) Let $m_k = \mathbb{E}X_k$. Show that $|m_k| \le \rho(h)^k|m_0|$ with $\rho(h) = \max_i|1 - h\lambda_i|$, with equality for suitable m_0. Show that the best rate is $\min_h \rho(h) = \frac{\kappa-1}{\kappa+1}$, where $\kappa = \lambda_{\max}/\lambda_{\min}$ is the condition number, and deduce that reducing the mean by a constant factor takes a number of iterations of order κ.

You may use that $\mathcal{W}_2(\mathcal{N}(a, s^2\mathbb{I}_n), \mathcal{N}(b, t^2\mathbb{I}_n))^2 = |a-b|^2 + n(s-t)^2$ and $\mathrm{KL}(\mathcal{N}(0, s^2\mathbb{I}_n)\|\mathcal{N}(0, t^2\mathbb{I}_n)) = \frac{n}{2}\left(\frac{s^2}{t^2} - 1 - \log\frac{s^2}{t^2}\right)$.

Exercise 2.2 (MALA satisfies detailed balance)**.** Let $\mu \propto e^{-U}$ with $U \in C^1(\mathbb{R}^n)$ and $\int e^{-U} < \infty$, and $h > 0$. The Metropolis-adjusted Langevin algorithm (MALA) proposes, from the current state x, a point y drawn from the ULA step $\mathcal{N}(x - h\nabla U(x), 2h\mathbb{I}_n)$, with density

$$p_h(x, y) = (4\pi h)^{-n/2} \exp\left(-\frac{|y - x + h\nabla U(x)|^2}{4h}\right),$$

and accepts it with probability

$$a(x, y) = \min\left\{1, \frac{\mu(y)p_h(y, x)}{\mu(x)p_h(x, y)}\right\}.$$

The MALA transition kernel is $P(x, \mathrm{d}y) = a(x, y)p_h(x, y)\,\mathrm{d}y + \rho(x)\delta_x(\mathrm{d}y)$, where $\rho(x) = 1 - \int a(x, y)p_h(x, y)\,\mathrm{d}y$ is the rejection probability (if rejected, the chain stays at x).

(a) Show that $\mu(x)a(x, y)p_h(x, y) = \min\{\mu(x)p_h(x, y), \mu(y)p_h(y, x)\}$, which is symmetric in (x, y).

(b) Deduce detailed balance: $\mu(\mathrm{d}x)P(x, \mathrm{d}y) = \mu(\mathrm{d}y)P(y, \mathrm{d}x)$ as measures on $\mathbb{R}^n \times \mathbb{R}^n$. Deduce that μ is invariant: $\int \mu(\mathrm{d}x)P(x, A) = \mu(A)$ for all Borel A.

(c) Write $\log\frac{\mu(y)p_h(y,x)}{\mu(x)p_h(x,y)}$ explicitly in terms of U, and conclude that MALA can be run without knowing the normalization constant of μ.

Exercise 2.3 (Score matching for Gaussian families)**.** Example 2.10 computes the score matching objective $J(m, A) := \mathbb{E}_\mu|s_{m,A}(X)|^2 + 2\mathbb{E}_\mu\nabla \cdot s_{m,A}(X)$ for the affine fields $s_{m,A}(x) = -A(x - m)$, and shows that $J(m, A) = \mathrm{tr}(A\Sigma A) - 2\,\mathrm{tr}\,A + |A(m_\mu - m)|^2$ for any μ with mean m_μ and covariance Σ.

(a) Given data $x_1, \dots, x_N \in \mathbb{R}^n$ with sample mean $\hat{m}$ and sample covariance $\hat{\Sigma} = \frac{1}{N}\sum_i (x_i - \hat{m})(x_i - \hat{m})^\top$, show that the empirical objective $\hat{J}_N(m, A) = \frac{1}{N}\sum_{i=1}^N \left(|s_{m,A}(x_i)|^2 + 2\nabla \cdot s_{m,A}(x_i)\right)$ has the unique minimiser $(\hat{m}, \hat{\Sigma}^{-1})$ when $\hat{\Sigma} \succ 0$, which is the Gaussian maximum likelihood estimator. Show that $\hat{J}_N$ is unbounded below when $\hat{\Sigma}$ is singular, in particular when $N \le n$.

(b) Restrict to isotropic fields, $A = a\mathbb{I}_n$ with $a > 0$. Show that the minimiser of J is $m = m_\mu$, $a = n/\operatorname{tr}\Sigma$, and that it coincides with the KL projection of μ onto the isotropic Gaussians $\{\mathcal{N}(m, \sigma^2\mathbb{I}_n)\}$.

(c) Restrict to diagonal $A = \operatorname{diag}(a_1, \dots, a_n)$. Show that the minimiser is $m = m_\mu$, $a_j = 1/\Sigma_{jj}$. Show that $a_j \le (\Sigma^{-1})_{jj}$, with equality if and only if the j-th coordinate is uncorrelated with the others under μ.

Exercise 2.4 (Denoising score matching and Tweedie's formula)**.** Let $X \sim \mu$ (any probability measure on $\mathbb{R}^n$) and $Z \sim \mathcal{N}(0, \sigma^2\mathbb{I}_n)$ be independent, $Y = X + Z$ and $\nu = \mathrm{Law}(Y)$.

(a) (Stein's identity.) Let $F \in C^1(\mathbb{R}^n; \mathbb{R}^n)$ with $\mathbb{E}|F(Z)|^2 < \infty$ and $\mathbb{E}|\nabla \cdot F(Z)| < \infty$. Show that $\mathbb{E}\,\nabla \cdot F(Z) = \sigma^{-2}\mathbb{E}\langle F(Z), Z\rangle$. *Hint:* let γ_σ be the density of Z and $G = F\gamma_\sigma$; show $G, \nabla \cdot G \in L^1$, then integrate $\nabla \cdot G$ against $\chi_R = \chi(\cdot/R)$, where $\chi \in C_c^\infty(\mathbb{R}^n)$, $0 \le \chi \le 1$, $\chi = 1$ on the unit ball, and let $R \to \infty$.

(b) (Tweedie's formula.) Show that ν has a positive C^∞ density and that

$$\nabla \log \nu(y) = -\frac{\mathbb{E}[Z \mid Y = y]}{\sigma^2} = \frac{\mathbb{E}[X \mid Y = y] - y}{\sigma^2}.$$

Deduce that $\mathbb{E}_\nu|\nabla \log \nu|^2 \le n/\sigma^2$.

(c) (Denoising score matching.) Let $s_\theta \in C^1(\mathbb{R}^n; \mathbb{R}^n)$ with $\mathbb{E}|s_\theta(Y)|^2 < \infty$ and $\mathbb{E}|\nabla \cdot s_\theta(Y)| < \infty$. Show that

$$\mathbb{E}|s_\theta(Y) - \nabla \log \nu(Y)|^2 = \mathbb{E}\left|s_\theta(Y) + \frac{Z}{\sigma^2}\right|^2 + C', \qquad C' = \mathbb{E}_\nu|\nabla \log \nu|^2 - \frac{n}{\sigma^2},$$

in two ways: (i) by first proving, with the cut-off argument of (a), the score matching identity $\mathbb{E}|s_\theta(Y) - \nabla \log \nu(Y)|^2 = \mathbb{E}|s_\theta(Y)|^2 + 2\mathbb{E}\,\nabla \cdot s_\theta(Y) + \mathbb{E}_\nu|\nabla \log \nu|^2$, and then using (a); (ii) from (b), by an orthogonal decomposition. Which assumption on s_θ does (ii) not need?

(d) Let $\mu = \mathcal{N}(0, a^2\mathbb{I}_n)$. Compute $\nabla \log \nu$, $\mathbb{E}[X \mid Y]$ and C', and discuss the bias $\nabla \log \nu - \nabla \log \mu$ and the minimal value $-C'$ of the denoising objective as $\sigma \to 0$.

Exercise 2.5 (Weak sensitivity to well-separated mixture weights)**.** For probability densities q, p on $\mathbb{R}^n$ that are C^1 and positive on the open set $\{q > 0\}$, the Fisher divergence is $\mathrm{FI}(q\|p) := \int_{\{q>0\}} q\,|\nabla \log q - \nabla \log p|^2\,\mathrm{d}x$. If q and p are positive and C^1 on all of $\mathbb{R}^n$, then $\mathrm{FI}(q\|p) = 0$ implies $q = p$ (Chapter 2). Fix a unit vector $e \in \mathbb{R}^n$. Only the coordinate $\langle e, x\rangle$, in the direction separating the modes, will matter.

(a) Let $\rho(x) = c\,e^{-1/(1-|x|^2)}$ for $|x| < 1$ and $\rho(x) = 0$ otherwise (a C^∞ probability density on $\mathbb{R}^n$, c the normalising constant), and for $w \in (0,1)$ let $q_w(x) = w\rho(x + 2e) + (1 - w)\rho(x - 2e)$. Show that $\nabla \log q_w$ does not depend on w on $\{q_w > 0\}$, hence the Fisher divergence (computed on $\{q_w > 0\}$) satisfies $\mathrm{FI}(q_w\|q_{w'}) = 0$ for all w, w', although $q_w \ne q_{w'}$ for $w \ne w'$. Why does this not contradict the implication $\mathrm{FI}(q\|p) = 0 \Rightarrow q = p$ stated above? What does the Langevin dynamics $\mathrm{d}X_t = \nabla \log q_w(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$ do when started at $X_0 = -2e$?

(b) Let $q_w = w\mathcal{N}(-me, \mathbb{I}_n) + (1-w)\mathcal{N}(me, \mathbb{I}_n)$, $m > 0$. Show that $\nabla \log q_w(x) = -x + m \tanh(m\langle e, x\rangle + c_w)\, e$ with $c_w = \frac{1}{2} \log \frac{1-w}{w}$.

(c) Show that $\mathrm{FI}(q_w \| q_{1/2}) > 0$ for $w \neq \frac{1}{2}$, but that $\mathrm{FI}(q_w \| q_{1/2}) \leq 16 c_w^2 e^{4|c_w|} m^2 e^{-2m^2} + 8m^2 e^{-m^2/8} \to 0$ as $m \to \infty$. *Hint:* $\tanh' = \mathrm{sech}^2 \leq 4e^{-2|\cdot|}$; split according to $|X| \leq m/2$ or not.